The forces at hand: social learning is permissive, not instructive, for the evolution of phenotypic plasticity

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Abstract

Evolution can favour a trait for two reasons: an instructive cause makes selection favour the trait; a permissive one lets it persist once selection no longer removes it. The distinction matters most for phenotypic plasticity and its replacement by inherited behaviour, genetic assimilation, the Baldwin effect. Social learning is credited with favouring plasticity on evidence that cannot tell the two apart: how much plasticity remains at the end of a simulation. Here we show, analytically and computationally, which role it plays in a minimal model of learning and evolution. Writing expected fitness as a function of plasticity, we find that environmental change is the only source of benefit from plasticity, and the social channel enters only its cost. Simulation confirms this: in a static environment selection against plasticity never reverses sign at any channel strength; when the environment changes, it reverses at a threshold the channel lowers, and without the channel the population collapses before reaching it. Social learning is permissive, not instructive: it supplies the means to be plastic, never the reason. For the debate over whether plasticity leads evolution, this is one case settled by measurement: environmental change directs, socially inherited information permits, and neither suffices alone.

Keywords: Baldwin effect; social learning; phenotypic plasticity; genetic assimilation; genetic drift; Rogers' paradox

1 Introduction

Phenotypic plasticity, the capacity of one genotype to produce different phenotypes in response to its environment, has moved to the centre of a debate about the structure of evolutionary theory [1, 2, 3, 4]. Proponents of an extended evolutionary synthesis argue that plasticity can lead adaptive evolution, with environmentally induced phenotypes appearing first and genetic change following, a sequence known as plasticity-led or plasticity-first evolution [5, 6]; critics reply that such cases are already accommodated by standard theory, in which selection on genetic variation remains the cause of adaptation [2]. Much of this disagreement is about causation [7]: whether plasticity helps to determine the direction of evolution, or merely allows evolution to proceed in a direction set by selection. Philosophers of biology call this the distinction between instructive and permissive causes [8, 9, 10]. Its sharpest case is genetic assimilation, the replacement of a learned or induced trait by an inherited one, and its best-known version, Baldwin's organic selection [11, 12, 13]: does learning steer assimilation, or does it only keep the population alive while selection does the steering?

The extended synthesis also widens what counts as inheritance, to include information transmitted socially rather than genetically [14, 3]. Social learning is therefore where its two central themes meet: it is a form of plasticity whose products are themselves inherited. Because its

effects on selection unfold over many generations, the question has been studied mainly in computational models, in which some aspects of an individual's phenotype are fixed by inheritance and others are left open, to be settled by learning during its lifetime [15, 16]. Learning can then guide a population to adaptations that genetic search alone would rarely find, and the amount of plasticity a population still carries at the end of a simulation became the standard evidence in these models that learning had been favoured by selection.

These models have since been extended to social learning, where an individual's learning is biased by information taken from others, and the extensions have repeatedly reported that this helps: Belew [17] found that a cultural bias on what an individual learns raises the amount of plasticity the population retains; Best [18] reported that copying a fitter individual's expressed behaviour can guide a population to its target with no individual learning at all; Le and colleagues [19, 20, 21] reported that a population of social learners sustains more plasticity, and reaches higher average fitness under a changing target, than individual learners alone; and in evolutionary robotics, controllers copied between robots are credited with a comparable adaptive advantage over evolution or individual learning on their own [22, 23, 24]. Read together, this body of work has settled into a common claim: that adding a social channel to this kind of model increases adaptation, and does so by increasing plasticity.

That claim rests on two things that have not, until now, been checked. First, on how the social channel is built. Most of the models above implement social learning as a dedicated procedure, separate from the individual learning it is compared against, which is precisely the assumption that Heyes [25] argues against on independent behavioural and comparative grounds: there is no dedicated social-learning mechanism, only one learning mechanism supplied with input that is tuned, developmentally or evolutionarily, towards other individuals. Section 2 reviews this literature in more detail and shows that the mismatch is the rule rather than the exception. Second, even where that mismatch is set aside, the claim rests on how much plasticity is retained, the same diagnostic that cannot distinguish an instructive cause from a permissive one. A population can end a run with more plasticity either because selection has come to favour staying open, or because the social channel has merely made staying open cheaper to carry; both produce the same number, and no study in this tradition has measured which one occurred.

Here we show, analytically and computationally, which role the social channel plays. We build a model in which social and individual information enter one learning rule, as Heyes's criterion requires, and we measure selection on plasticity directly, as the regression of an individual's fitness on its own plasticity, rather than reading it off a final frequency. Written as a function of plasticity, expected fitness has a single term that rewards plasticity, and it belongs to the environment; the social channel enters only the cost. The channel is therefore permissive, not instructive. A sweep over channel strength and rate of environmental change confirms this, and the same instruments resolve a twenty-year disagreement about whether social learning alone can guide a population to its target (Appendix A).

The remainder of the paper is organised as follows. Section 2 reviews how social learning has been implemented and why it matters; Section 3 presents the model, the measurement and the analytic criterion; Section 4 reports the results; and Section 5 returns to the debate with which we began.

2 Related work

Whether phenotypic plasticity can lead adaptive evolution is one of the questions on which the proposed extension of the evolutionary synthesis turns [1, 2, 3, 4]. In the plasticity-led account a new environment first induces a phenotype, the induced phenotype exposes previously cryptic genetic variation, and selection then refines and eventually fixes it, so that plasticity precedes and

directs the genetic change [5, 6]. The standard account replies that nothing in this departs from selection acting on genetic variation: induction may create the opportunity, but the direction of evolution is still set by which variants selection favours [2, 26]. The oldest version of the dispute concerns genetic assimilation and Baldwin's organic selection [11, 12], and even its modelling is unsettled: Scheiner, Barfield and Holt [13] found that assimilation as originally proposed is too slow to be likely, while plasticity itself can rescue populations from extinction under environmental change. What both sides ask for is the same thing. Direct measurement of selection on plasticity is among the criteria Levis and Pfennig [5] set for a plasticity-led case, and it is rarely available in nature; the computational models reviewed later in this section show that it has rarely been attempted even where it is cheap.

Underneath the dispute is a question about causal role. Developmental biology distinguishes an instructive signal, which determines which of several outcomes a competent tissue produces, from a permissive one, which the tissue needs in order to produce an outcome already determined by something else [10]. Woodward [27] gave the distinction a general form: a cause is specific to an effect to the extent that interventions on it map onto distinct values of the effect. Calcott [8] showed that instructive causes are the specific ones and permissive causes the unspecific, and Bourrat [9] examined the limits of the contrast. Uller and Laland [7] argue that much of the extended-synthesis debate is of this kind: both sides agree that development, inheritance and environment are causally involved in evolution, and disagree about which of them specify its direction. Put this way the question has a measurable form. A cause whose intervention changes the sign of selection on a trait is instructive with respect to that trait; one whose intervention changes only the strength of selection, or whether selection can act at all, is permissive. This paper builds a model in which both kinds of intervention can be made, on the environment and on the social channel, and selection on plasticity is measured after each.

The extended synthesis also widens inheritance to include information transmitted socially [14, 3], which makes social learning the case in which plasticity and inheritance meet. Its evolution has a mature theory. Boyd and Richerson [28] formalised the biases by which individuals choose what to copy, including the conformity bias that the second parameter of our social channel borrows; Rogers [29] showed that social learners can invade a population without raising its mean fitness, because they consume information that others produce; Borenstein, Feldman and Aoki [30] and Ram, Liberman and Feldman [31] showed that which mode of learning selection favours depends on how the environment fluctuates, and that the stable rate of a transmission mode can differ from the rate that maximises population fitness; Aoki and Feldman [32] review the field. These models are analytic and precise, but their variable is the frequency of a learning strategy, not selection on plasticity within a genome, so they cannot say whether a social channel instructs or permits. On mechanism, comparative psychology and neuroscience agree: social and asocial learning use the same learning process, and what is distinctively social is the input, perceptual and attentional processes tuned towards other individuals [25, 33, 34]. This paper adopts that architecture, one learning rule with a heritable social input, which is what allows selection on the input itself to be measured.

Models of learning and evolution supply the setting in which plasticity and its assimilation can be counted exactly. Hinton and Nowlan [15] (henceforth H&N) showed that lifetime learning lets a population find a target that genetic search alone would not, and the model has been analysed repeatedly: Fontanari and Meir [35] and Belew [17] derived its waiting times, Harvey [36] showed that the plasticity it retains is maintained by drift rather than by selection, Ancel [37] showed in a quantitative-genetic version that plasticity can even slow adaptation, Mayley [38] added a cost of learning, and Morgan et al. [39] and Santos et al. [40] warned that what the effect shows depends on the kind of plasticity assumed. Nolfi and colleagues reproduced the effect in neural networks and noted that it raised average rather than peak fitness [16, 41]. Social learning was added in three forms that differ in where the social signal enters the learner: as the learner's training target [42, 43, 44], as a bias on the learner's own guesses [17], or as a

value copied directly into its undetermined loci, bypassing learning [18, 19, 20, 21, 22, 23, 24]; recent neural-network [45] and reinforcement-learning [46] models use the first two. In every case the evidence that social learning favours plasticity is the plasticity remaining at the end of a run, and the tradition carries an unresolved contradiction, between Best's [18] report that social learning alone can guide a population to its target and Le's [19] that it cannot, a contradiction resolved, with the present model in hand, in Appendix A and taken up in the Discussion.

The instruments the question needs already exist. Lande and Arnold [47] showed that selection on a trait can be measured as the regression of relative fitness on the trait within a generation, with no assumption about inheritance, and the method has since been applied to thousands of traits in natural populations [48]; it has not been applied to plasticity in the models above. Harvey [36] supplied the complementary test for drift on this very landscape, the distribution of per-locus frequencies, which separates a trait held at an intermediate frequency by selection from one that persists because selection has stopped. With a selection gradient on plasticity, a convergence test and a drift test applied to one model in which plasticity is a count, the causal question above becomes answerable, and the next section builds that model.

3 Model

The model has three parts: a population whose members carry some traits fixed by inheritance and leave others to be learned; a social channel through which the learning of one individual is informed by others; and an environment that may change. We define each in turn, then derive what the model implies about selection on plasticity before any simulation is run, and finally describe how selection is measured in simulation.

3.1 Choice of landscape

We place the model on the landscape of H&N for three reasons. First, plasticity is defined exactly on it: each locus is either fixed by inheritance or left open, so an individual's plasticity is a single count and selection on it can be measured without ambiguity. Second, the landscape has no fitness gradient other than the one learning creates, so any selection on plasticity that we measure is due to learning and the environment, not to the structure of the fitness function. Third, it is the landscape on which the computational claims reviewed in Section 2 were made, so a result on it speaks to those claims directly.

3.2 Population and lifetime learning

A population of N haploid individuals is followed in discrete, non-overlapping generations. Each individual i carries L loci, and the allele x_{ij} at locus j takes one of three values: 0 or 1, which are *determined* by inheritance, or "?", which is *undetermined* and must be set by learning. The environment rewards a single target string $\theta = (\theta_1, \ldots, \theta_L) \in \{0,1\}^L$. We call the number of undetermined loci of an individual its *plasticity*,

$$n_{?,i} \;=\; \#\{\, j : x_{ij} = ?\,\},$$

and write q for a particular value of $n_?$. The population frequency of undetermined alleles, the mean of $n_{?,i}/L$ over individuals, is written $p(?)$ throughout.

An individual with a determined allele that differs from the target at any locus cannot express the target and receives the minimum fitness $F_{\min} = 1$. An individual whose determined alleles all match the target learns by trial and error: in each of up to G trials it sets every undetermined locus at random, as described below, and it stops at the first trial g on which all of them match

the target. Its fitness is

$$F \;=\; 1 + 19\,\frac{G - \min(g, G)}{G}, \tag{1}$$

which runs from $F_{\min} = 1$ (no success within G trials) to $F_{\max} = 20$ (an individual with no undetermined loci, which succeeds without learning). The earlier an individual learns the target, the fitter it is, so undetermined loci carry a cost: each one is another locus that must be guessed correctly.

Parents are drawn with probability proportional to fitness, and offspring are formed by single-point crossover of two parents. There is no mutation at the loci, so the only sources of change in the genetic composition of the population are selection and drift.

3.3 The social channel

Learning is informed by the previous generation. Let the *demonstrators* be the individuals of the previous generation that reached the target ($F > F_{\min}$). A learner observes the phenotype of k demonstrators, sampled with replacement; at each locus a demonstrator expresses the target value, because it reached the target. Each observed value is perceived correctly with probability f, the *fidelity* of transmission, and as the opposite value otherwise. The *social cue* at locus j, $c_j \in [0, 1]$, is the fraction of the k perceived values that equal 1.

The learner combines the cue with its own uninformed guess in a single learning rule. Its probability of setting locus j to 1 on a trial is

$$\pi_j \;=\; D_b(\hat{c}_j), \qquad \hat{c}_j \;=\; a\,c_j + (1 - a)\,\tfrac{1}{2}, \tag{2}$$

where

$$D_b(x) \;=\; x + b\,x(1 - x)(2x - 1). \tag{3}$$

The two parameters correspond to the two parts of the architecture of Section 2. The *attention gate* $a \in [0, 1]$ sets how much social information enters the learning rule: at $a = 0$ the cue is ignored and every guess is a fair coin, which is the asocial model of H&N; at $a = 1$ the learner guesses in proportion to the cue. The *conformity transform* D_b, with $b \in [-1, 1]$, sets how the admitted information is processed. It fixes the points 0, $\frac{1}{2}$ and 1 and satisfies $D_b(1 - x) = 1 - D_b(x)$; $b > 0$ pushes a cue towards the majority value, $b < 0$ pulls it towards $\frac{1}{2}$, and $b = 0$ uses it as it is. The social channel therefore changes the input to learning, not the learning rule: guessing is the same process with or without it.

How demonstrators are sampled turns out to matter. Unless stated otherwise, the $k = 10$ demonstrators are drawn uniformly from all those of the previous generation. A second rule is used in Sections 4.3 and 4.4: $k = 5$ demonstrators drawn from the fittest tenth of the previous generation, so that the cue is itself a sample that varies from one generation to the next. A third, a single demonstrator that is the fitter of the learner's two parents, so that the cue is inherited along a lineage and carries no variance of its own, is used in Appendix A.

In most experiments a and b are set by the experimenter and are the same for every individual, so that they can be intervened on. In Section 4.4, a is instead a heritable trait: each offspring inherits the value of one parent with a Gaussian perturbation, so that selection on the channel itself can be observed.

3.4 Environmental change

Each generation, with probability ψ, one locus of the target, chosen at random, is flipped. The rate ψ is the environmental parameter of the model; $\psi = 0$ is a static environment. A flip turns every individual that carries a determined allele at that locus from a match into a mismatch, and leaves individuals for whom the locus is undetermined unaffected, since they set it afresh

by learning. Because the loci do not mutate, a population in which no individual any longer reaches the target cannot recover it; we say such a population has *collapsed*, and write Pr(live) for the fraction of runs that still contain target-reaching individuals (more than one-half of a winner per generation, averaged over the final fifty generations). A collapsed population persists at the fitness floor; nothing demographic is implied.

3.5 Analysis: where the environment and the channel enter selection

We now derive, for this model, which parameters can change the direction of selection on plasticity and which can only change its strength. The derivation uses a mean-field approximation: the cue is replaced by its expectation, and a single generation is considered.

Probability of learning the target. Because a demonstrator expresses the target value at every locus, the expected fraction of perceived values that match the target is f. By the symmetry $D_b(1-x) = 1 - D_b(x)$, the probability that a learner sets locus j to its target value is then the same whether $\theta_j = 0$ or 1:

$$p \;=\; D_b\big(af + (1-a)\,\tfrac{1}{2}\big). \tag{4}$$

Guesses at different loci and on different trials are independent, so an individual whose q undetermined loci are the only ones it must learn succeeds on a given trial with probability $P = p^{\,q}$, and the number of trials it uses is geometric, truncated at G:

$$E\big[\min(g, G)\big] \;=\; \sum_{t=0}^{G-1} (1-P)^t \;=\; \frac{1-(1-P)^G}{P}. \tag{5}$$

By equation (1), the expected fitness of an individual whose determined alleles all match the target is

$$F_{\text{learn}}(q) \;=\; 1 + 19\left(1 - \frac{1-(1-p^q)^G}{G\,p^q}\right). \tag{6}$$

This depends on a and b only through p, and it does not depend on ψ.

Selection on plasticity. We measure selection on plasticity as the regression of fitness on plasticity across the individuals of a generation [47],

$$\beta_q \;=\; \frac{\operatorname{Cov}(F, n_?)}{\operatorname{Var}(n_?)}. \tag{7}$$

Since $\operatorname{Cov}(F, n_?) = \operatorname{Cov}(E[F \mid n_?], n_?)$, β_q is the regression of the conditional mean fitness $E[F \mid q]$ on q. If $E[F \mid q]$ changes in the same direction at every value of q present in the population, β_q has the sign of $\Delta(q) = E[F \mid q+1] - E[F \mid q]$, the change in expected fitness when one more locus is left undetermined. Positive β_q means that selection favours plasticity, negative that it removes it, and zero that it is neutral.

The static environment. When $\psi = 0$, and once determined alleles that do not match the target have been removed, every individual's expected fitness is $F_{\text{learn}}(n_?)$.

Proposition 1 *If $\psi = 0$ and all determined alleles match the target, then $\Delta(q) \le 0$ for every q and every value of a, b and f, with equality only if $p = 1$.*

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Proof. Leaving one more locus undetermined multiplies the per-trial success probability by $p \leq 1$, from p^q to p^{q+1}. The expected number of trials in equation (5) is a sum of terms $(1-P)^t$, each non-increasing in P, so it does not decrease; by equation (1), expected fitness does not increase. If $p < 1$ the decrease is strict. □

The social channel enters this result only by raising p towards 1. It can therefore make the cost of plasticity arbitrarily small, but it cannot make plasticity advantageous. Before mismatching determined alleles have been removed, an undetermined locus can be favoured, because a determined one may be wrong; this is the classical Baldwin effect on this landscape, and it is transient in a static environment. The next case shows that a changing environment makes the same mechanism permanent.

The changing environment. When $\psi > 0$, a determined allele can become a mismatch. With one locus flipped per change event, an individual with q undetermined loci has a determined allele at the flipped locus with probability $(L-q)/L$, so its target is spoiled in a given generation with probability $\psi(L-q)/L$, after which its fitness is $F_{\min}$. Its expected fitness is therefore

$$E[F \mid q] \;=\; \Big(1 - \psi\,\frac{L-q}{L}\Big)\,F_{\text{learn}}(q) \;+\; \psi\,\frac{L-q}{L}\,F_{\min}. \tag{8}$$

Write $C(q) = F_{\text{learn}}(q) - F_{\text{learn}}(q+1) \geq 0$ for the cost of one more undetermined locus. Subtracting equation (8) at q from its value at $q+1$ and collecting terms gives

$$\Delta(q) \;=\; \underbrace{\frac{\psi}{L}\,\big(F_{\text{learn}}(q+1) - F_{\min}\big)}_{\text{benefit}} \;-\; \underbrace{\Big(1 - \psi\,\frac{L-q}{L}\Big)\,C(q)}_{\text{cost}}. \tag{9}$$

Proposition 2 *The benefit of plasticity in equation (9) is proportional to the rate of environmental change ψ and is zero when $\psi = 0$. The parameters of the social channel, a and b, enter $\Delta(q)$ only through p, and therefore only through F_{learn} and the cost $C(q)$. Selection favours plasticity at q if and only if*

$$\psi \;\geq\; \psi^*(q) \;=\; \frac{L\,C(q)}{F_{\text{learn}}(q+1) - F_{\min} + (L-q)\,C(q)}. \tag{10}$$

Proof. The first two statements follow from equations (4)–(8), in which ψ appears only in the probability of being spoiled and a, b only in p. Setting $\Delta(q) \geq 0$ in equation (9) and solving for ψ gives equation (10). □

Numerically, $\psi^*(q)$ decreases monotonically as p increases, at every q from 1 to $L-1$ (for $L = 20$, $G = 1000$), and tends to zero as $p \to 1$, because the cost $C(q)$ vanishes. A more open channel therefore lowers the rate of change at which plasticity becomes favoured.

Equation (8) also idealises the cue: in simulation the cue is built from the previous generation's winners, so in the generation after a flip it points at the old value of the flipped locus, a cost of relying on the channel that grows with ψ and that the derivation omits. The omission biases the predicted threshold downwards, and consistently, every observed crossing in Section 4.2 lies at or above its prediction.

Causal roles. Equation (9) assigns the two parameters different jobs, and we read their causal roles off that decomposition, in the sense of causal specificity [27, 8] applied term by term. The environment is the *instructive* cause of selection for plasticity: the benefit term exists only through ψ and vanishes identically at $\psi = 0$ (Proposition 1), whatever the channel does. The channel is *permissive*: a and b enter only the cost term, so opening the channel rescales the penalty and relocates the threshold ψ^* (Proposition 2), but cannot create a benefit where the

environment supplies none. The asymmetry is between the terms, not between which knob can flip an observed sign: at a fixed $\psi > 0$, opening the channel can tip Δ from negative to positive by cheapening plasticity, just as removing a constraint can expose a selection pressure it was masking. What no setting of the channel can do is make plasticity favoured in a world that never changes. The channel also matters in a way equation (9) does not capture: whether a population survives repeated changes long enough for selection to act at all. That is a population-level property, and it is measured in Section 4.

The analysis is a one-generation, mean-field approximation evaluated at a single value of q. It predicts the sign of selection where $\Delta(q)$ is clearly positive or negative across the population, and may fail near the threshold, where the population spans values of q on both sides. Whether it predicts the simulation is tested in Section 4.2.

3.6 Measurement in simulation

Selection on plasticity is computed in every generation as β_q in equation (7), in units of fitness per undetermined locus. Two summaries are reported. The first averages β_q over the final generations of a run. Under a changing environment plasticity can approach fixation, so that $\mathrm{Var}(n_?)$ becomes small and β_q noisy; we then also report $\beta_q \cdot \mathrm{SD}(n_?)$, the expected difference in fitness between individuals one standard deviation apart in plasticity, and use a second, more conservative summary: the average of β_q over every generation after a burn-in of 50 in which $\mathrm{Var}(n_?) \geq 0.05$. Each surviving run then contributes one estimate, and we report how many runs give a positive estimate, with an exact two-sided binomial test against one half. All values of β_q are conditional on the population retaining the target; runs that fix completely at "?" contribute no end-of-run estimate, and the run counts reported alongside each cell carry the between-run variation. With 84 cells in the sweep we treat individual cell tests as descriptive; the only cell-level result claimed as confirmatory is the unanimity of the $\psi = 0$ column, whose per-cell $p = 4.9 \times 10^{-4}$ survives a Bonferroni correction across the grid. When a is heritable (Section 4.4), the same regression with a in place of $n_?$ measures selection on the channel itself. All simulations reported here were run with NumPy 2.4.3 under fixed random seeds; random streams differ between NumPy versions, so exact reproduction requires the archived environment.

Two further tests separate selection from drift. In a *convergence test*, runs are started from different initial levels of plasticity; if selection holds plasticity at an equilibrium, the runs converge on it, and if it does not, they retain their initial differences. In a *locus frequency test* [36], we record the frequency of "?" at each locus at the end of long runs: a level of plasticity maintained by selection appears as loci held at intermediate frequencies, whereas plasticity that persists because selection no longer removes it appears as loci fixed at either end, with the population mean equal to the fraction of loci fixed at "?".

4 Results

Unless stated otherwise: $N = 1000$, $L = 20$, $G = 1000$, 400 generations, $k = 10$ demonstrators drawn uniformly from those of the previous generation, fidelity $f = 0.9$, a and b fixed and equal for all individuals, one locus flipped per change event, and 12 independent runs per condition; in the sweep of Section 4.2 one demonstrator sample is drawn per generation and shared by all learners. Reported values are means over the final generations of each run (the convergence runs of Table 2 last 600 generations). Sections 4.3 and 4.4 use a second implementation of the same model: learning budget $G = 300$, $k = 5$ demonstrators sampled per learner from the fittest tenth of the previous generation (winners or not; a non-winner displays its current guesses at undetermined loci), perfect transmission ($f = 1$), $b = 0$, and initial $p(?) = 0.5$.

Absolute fitness values are therefore not comparable between the two implementations, and no cross-implementation comparison is made. Each subsection asks one question of the analysis of Section 3.5 and reports the measurement that answers it.

4.1 Does any setting of the channel make selection favour plasticity in a static environment?

Proposition 1 says no: at $\psi = 0$, once mismatched determined alleles are gone, one more undetermined locus can only slow learning, whatever the channel admits or how it processes it. What the analysis does not say is how much of the penalty the channel removes, and what the plasticity that escapes removal then does; those are measurements. Table 1 gives β_q and the final frequency of undetermined alleles for three configurations, each started from a low and a high initial level of plasticity.

Table 1: Selection on plasticity in a static environment ($\psi = 0$). β_q is fitness per plastic allele, equation (7).

	initial $p(?) = 0.2$		initial $p(?) = 0.8$	
configuration	β_q	final $p(?)$	β_q	final $p(?)$
$a = 0$ (asocial, original model)	−0.2106	0.1661	−0.1132	0.1362
$a = 0.8$, $b = 0$ (attention only)	−0.0167	0.3319	−0.0625	0.6350
$a = 0.8$, $b = 0.5$ (attention and conformity)	−0.0085	0.3271	−0.0386	0.8001

β_q is negative in every cell. Opening the attention gate weakens the penalty on plasticity by a factor of 12.6, and adding conformity weakens it by 24.8, but neither changes its sign. On the standard diagnostic the same runs would look like a strong effect: the final frequency of undetermined alleles rises from 0.17 to 0.33 and, from a high start, to 0.80. Read through β_q, nothing has been selected for. The social channel has made plasticity cheap, not advantageous.

If the weakened penalty of Table 1 were instead a selective equilibrium at a higher level of plasticity, runs started at different levels would converge on it. Table 2 starts three runs at $p(?) = 0.2$, 0.5 and 0.8, an initial spread of 0.60, and reports what remains.

Table 2: Convergence test. Final spread is the range across the three starting points; the last column expresses it as a fraction of the initial spread of 0.60.

configuration	from 0.2	from 0.5	from 0.8	final spread	initial spread retained
$a = 0$ (asocial)	0.1591	0.1008	0.1199	0.0583	9.7%
$a = 0.8$	0.3332	0.4820	0.5663	0.2331	38.9%
$a = 1.0$	0.3047	0.5769	0.8321	0.5274	87.9%

The asocial model erases ninety per cent of its initial condition: selection is working and there is an equilibrium to find. At $a = 0.8$ the spread is still closing, from 0.48 at generation 100 to 0.23 at 600, at a pace consistent with selection weakened twenty-five-fold rather than removed; a fixed-horizon spread cannot by itself distinguish neutralised from slowed selection. At $a = 1.0$ nothing closes, and the final level of plasticity is a memory of the initial condition. The locus-frequency test of Section 4.3 makes the distinction cleanly.

The answer to the question of this subsection is therefore no: selection against plasticity keeps its sign at every opening of the channel; the channel changes how fast, and ultimately whether, that selection still moves the population.

The sweep of Section 4.2 closes the remaining gap: in its $\psi = 0$ column, $\beta_q < 0$ in every one of its 132 distinct runs (b has no effect at $a = 0$, so two cells coincide), so the answer does not change at any opening of the channel.

4.2 What reverses the sign of selection: the channel or the environment?

Figure 1 sweeps the attention gate from $a = 0$ to 1 and the rate of change from $\psi = 0$ to 0.25, for $b = 0$ and $b = 0.5$, twelve independent runs per cell, with the trajectory estimator of Section 3.6. Proposition 2 makes three predictions about this plane: the column $\psi = 0$ is negative at every a and b; for an open channel the sign crosses at $\psi^*(\bar{q})$ of equation (10); and the crossing moves to smaller ψ as the channel opens, because the channel sets only the cost term. The figure shows all three, and one thing the proposition is silent about.

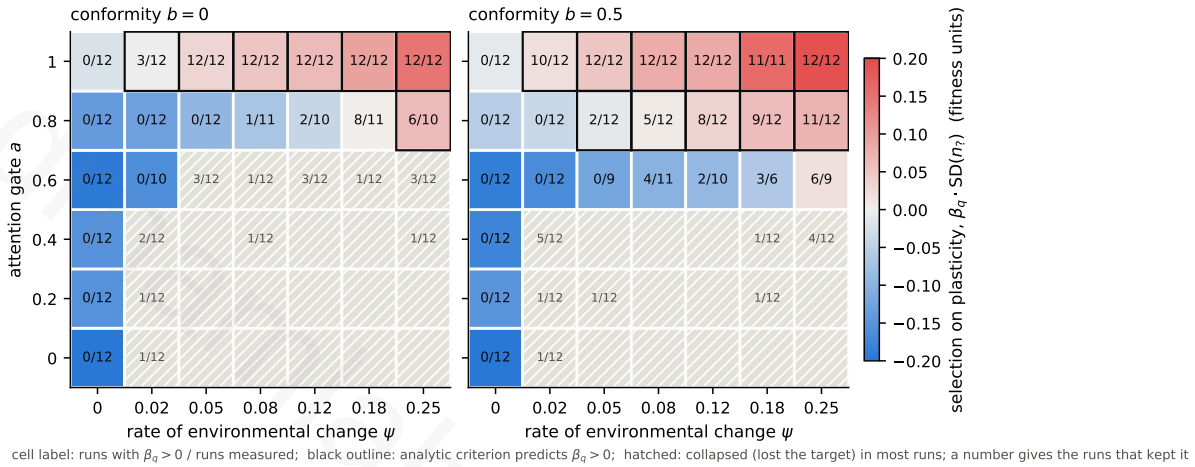


Figure 1: Sign of selection on plasticity across channel strength a and rate of environmental change ψ, for conformity $b = 0$ (left) and $b = 0.5$ (right); 12 independent runs per cell, initial $p(?) = 0.8$. Cell colour is the mean of $\beta_q \cdot \mathrm{SD}(n_?)$ over runs, measured along the trajectory; the label is the number of runs with $\beta_q > 0$ over the number measured. Hatched cells are conditions in which the population collapsed (lost the target) in more than half of the runs. A black outline marks cells in which equation (9), evaluated at the cell's own mean plasticity, predicts $\beta_q > 0$.

First, the column $\psi = 0$ is negative at every level of the channel. In all 132 distinct runs of that column (b is inert at $a = 0$, so two cells coincide), $\beta_q < 0$, with an exact binomial $p \leq 0.001$ in every cell; the cell means run from -0.237 at $a = 0$ to -0.007 at $a = 1$, $b = 0.5$. This is Proposition 1 again, now at every level of the channel rather than at the three of Table 1.

Second, and this is the part equation (9) does not contain, below $a = 0.6$ populations with $\psi \geq 0.05$ collapse in at least eight of twelve runs, and in most of those conditions in all twelve; at $a = 0.6$ survival is partial and the majority of runs stays non-positive below $\psi = 0.25$. The channel decides whether there is a population left to measure.

Third, for $a \geq 0.8$ the sign reverses with ψ, and the crossing moves to lower ψ as the channel opens. At $a = 1$ with $b = 0$, $\beta_q > 0$ in twelve of twelve runs from $\psi = 0.05$ ($p < 0.001$); with $b = 0.5$ the majority is already positive at $\psi = 0.02$ (ten of twelve, $p = 0.039$) and unanimous from $\psi = 0.05$; at $a = 0.8$ the majority of runs turns positive between $\psi = 0.12$ and 0.18 ($b = 0$) and between 0.08 and 0.12 ($b = 0.5$); at $a = 0.6$ it does not turn positive below $\psi = 0.25$ for $b = 0$ and does so only at $\psi = 0.25$ for $b = 0.5$.

Table 3 compares these crossings with $\psi^*(\bar{q})$ from equation (10). The predicted threshold lies within or adjacent to the observed bracket in five of the six configurations and one grid step early in the sixth ($a = 0.8$, $b = 0.5$). Cell by cell, the sign of $\Delta(\bar{q})$ in equation (9) at the cell's own $\bar{q}$ matches the majority sign of β_q in 38 of the 43 cells with at least six measured runs, and all five mismatches lie on the boundary, where the fraction of positive runs is between 2/12 and 8/11. The end-of-run estimate of Table 4 put the crossing for $a = 0.8$, $b = 0.5$ at $\psi \approx 0.06$; the trajectory estimate is the more conservative of the two and lies in the same region.

Table 3: Predicted and observed thresholds. p is the per-locus probability of guessing correctly at a cue of accuracy $f = 0.9$; $\bar{q}$ is the mean number of undetermined loci in the $\psi = 0.05$ cell, at which ψ^* from equation (10) is evaluated; the observed bracket is the pair of adjacent grid values between which the majority sign of β_q changes.

a	b	p	$\bar{q}$	predicted ψ^*	observed crossing
0.6	0	0.74	12	0.247	none below 0.25
0.8	0	0.82	16	0.106	0.12–0.18
1.0	0	0.90	18	0.015	0.02–0.05
0.6	0.5	0.79	15	0.200	0.18–0.25
0.8	0.5	0.87	18	0.040	0.08–0.12
1.0	0.5	0.94	19	0.005	0–0.02

The three features are what Propositions 1 and 2 say. The environment is the only term that rewards plasticity, so without environmental change the channel can only weaken the penalty; the channel is the only term that prices plasticity, so it sets both the threshold at which change makes plasticity worth having and whether the population survives to reach it, without ever supplying the direction.

A finer slice through this plane, at the single setting $a = 0.8$, $b = 0.5$ and with the end-of-run estimator, shows the crossing itself. Table 4 sweeps the rate of environmental change at the configuration that was most permissive in the static case.

Table 4: Moving target, $a = 0.8$, $b = 0.5$, initial $p(?) = 0.8$. Pr(live) is the fraction of runs whose population survives to the end. The last column is β_q multiplied by the standard deviation of $n_?$, that is, the expected fitness difference for a one-standard-deviation difference in plasticity, on a scale from 1 to 20.

ψ	Pr(live)	$p(?)$	β_q	Var($n_?$)	β_q·SD
0.00	1.00	0.8001	−0.0386	1.892	−0.0531
0.02	1.00	0.8349	−0.0188	1.540	−0.0233
0.05	1.00	0.9132	−0.0163	0.825	−0.0148
0.08	1.00	0.9259	**+0.0204**	0.688	+0.0169
0.12	1.00	0.9543	+0.1877	0.358	+0.1123
0.18	1.00	0.9850	+1.6526	0.121	+0.5739
0.25	1.00	0.9900	+1.3071	0.078	+0.3640

Linear interpolation puts the sign change at $\psi^* = 0.0633$. The obvious objection is that Var($n_?$) falls by a factor of twenty-four across the sweep, so a large β_q could be a small denominator. The final column answers it: $\beta_q \cdot \mathrm{SD}(n_?)$ changes sign at the same place and reaches +0.574, more than half a fitness unit, so the reversal is not an artefact of scaling.

The same slice with the channel closed shows what the channel's survival role means in practice. Table 5 repeats the sweep with the attention gate closed.

There is no regime in which plasticity is positively selected, because the population does not survive long enough to have one. Survival has fallen to a quarter of runs by $\psi = 0.005$, and wherever a coefficient can be computed it is not merely negative but more negative than in the static case. The regime identified in Table 4 is therefore not a property of environmental change; it is a property of environmental change *together with* a social channel.

Table 6 shows that both components are required, and that the initial level of plasticity is a third requirement.

With the channel closed the population collapses whatever its plasticity. With the channel open but little plasticity to begin with, it also collapses. Survival requires the peripheral gate

Table 5: Control: the same sweep with $a = 0$. Dashes indicate that no run retained the target, so no coefficient is defined. Bold in Table 4 marks the first row with $\beta_q > 0$.

ψ	Pr(live)	$p(?)$	β_q	$\beta_q \cdot$SD
0.000	1.00	0.1362	−0.1132	−0.0818
0.005	0.25	0.1781	−0.2022	−0.1619
0.010	0.00	–	–	–
0.020	0.08	0.2004	−0.2886	−0.0324
0.030	0.00	–	–	–
0.050	0.00	–	–	–

Table 6: Both channels needed, at $\psi = 0.05$. Winner is the number of individuals out of 1000 expressing the target in the final generation.

a	b	initial $p(?)$	Pr(live)	$p(?) \mid$ live	Winner
0.0	0.0	0.2	0.00	–	0
0.0	0.0	0.8	0.00	–	0
0.8	0.0	0.2	0.00	–	0
0.8	0.0	0.8	1.00	0.8022	990
0.8	0.5	0.2	0.08	0.5164	51
0.8	0.5	0.8	1.00	0.9132	993

open *and* a stock of plasticity for the incoming signal to act on, and conformity, the central parameter, rescues a small fraction of the otherwise fatal low-plasticity condition rather than substituting for either.

4.3 Is the plasticity the channel preserves held by selection, or by drift?

Proposition 1 leaves only one candidate for what maintains the plasticity that an open channel preserves in a static environment: since no interior level of plasticity is an optimum, a population can hold one only by drift. Drift makes three predictions that selection does not. Individual loci should fix at one end or the other, rather than sit at a shared intermediate frequency; the population mean should equal the fraction of loci that happen to have fixed at "?"; and the retained level should shrink as the population grows. The locus-frequency test of Section 3.6 measures the first two directly: a selective equilibrium at an intermediate allele frequency should show up as a frequency at which loci actually sit, not only as a stable population mean, which can equally be produced by a population split between loci fixed at either end. Figure 2 reports the result, and Table 7 collects the three fractions.

Table 7: Locus-frequency distribution after $T = 3000$ generations, six runs × 20 loci, a evolving, $k = 5$, fittest tenth. Bold marks the stuck class, whose fraction equals the population frequency of "?".

loci fixed to a determined value (frequency < 0.01)	0.725
loci **stuck** at "?" (frequency > 0.99)	**0.275**
loci at an intermediate frequency (0.15–0.85)	0.000

The distribution is bimodal: no locus sits at an intermediate frequency after three thousand generations, which is not what a selective equilibrium looks like and is exactly what drift looks like. The stuck-locus fraction, 0.275, equals the population-level "?" frequency of the same runs; the same configuration stood at 0.327 at generation 2500 (six runs) and is still drifting towards fixation at 3000. This is Harvey's identity, that the population mean under drift equals the fraction of loci that happen to have fixed in the undetermined state, not a frequency that any

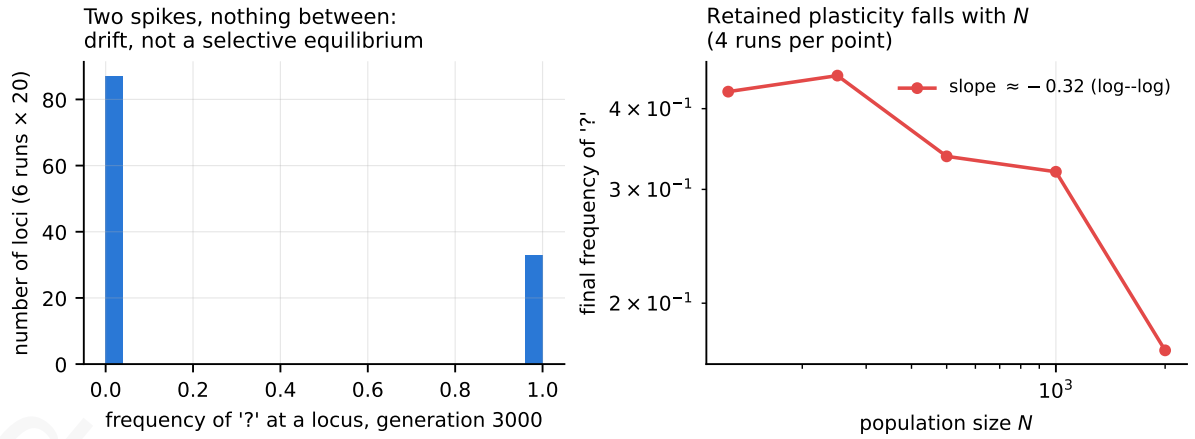


Figure 2: Left: the distribution, across loci and six long runs, of the frequency at which each locus sits after 3000 generations, with a evolving, $k = 5$, fittest tenth. The distribution is two spikes at the ends, with essentially nothing between. Right: the final population frequency of "?" falls with population size N, consistent with a drift process rather than a fixed equilibrium.

single locus is held at by selection. Best [18], looking at the same persistence a quarter-century ago, judged that "the forces at hand are genetic drift ... versus this selection pressure towards assimilation"; the measurements above give that sentence its numbers, and this paper its title. The final "?"-frequency also falls as N grows across a sweep from $N = 125$ to $N = 2000$ (noisy at the smaller sizes, clearer for $N \geq 500$), the signature of a quantity shrinking under drift as the population gets larger, which a selected equilibrium frequency would not do.

All three predictions of drift are met, and none of selection. The social channel does not reward plasticity, it relaxes the penalty on it, and genetic drift does the rest, settling at a higher stuck fraction than the asocial model's 0.094–0.100 because the penalty it is drifting against is weaker. We have not found a configuration, across Sections 4.1–4.3, in which an open social channel on a static landscape produces a selective reward for plasticity; every case we can measure is consistent with permission rather than instruction, now by three independent instruments – the regression β_q, the convergence test, and Harvey's own locus-frequency diagnostic.

4.4 If the channel is permissive, why does selection open it?

Everything so far holds the channel fixed. Equation (9), however, is a statement about a single individual: whatever ψ is, raising a lowers that individual's cost of plasticity, so wherever plasticity exists, an individual with a more open gate should outperform one with a less open gate, whether or not the population as a whole gains anything. The analysis therefore predicts a gap between the two levels of selection, which is Rogers' [29] structure. Letting a evolve as a heritable trait (Section 3.6) makes both levels measurable at once (figure 3).

Table 8: Individual versus population: 400 generations, $k = 5$, fittest decile, a evolving with Gaussian mutation, last 100 generations averaged.

correlation between individual a and individual fitness	+0.491
mean fitness, top quartile of a	19.930
mean fitness, bottom quartile of a	19.694
mean evolved a	0.830
mean fitness of the population that evolves a freely	19.848

Within the population, the correlation between an individual's own a and its own fitness

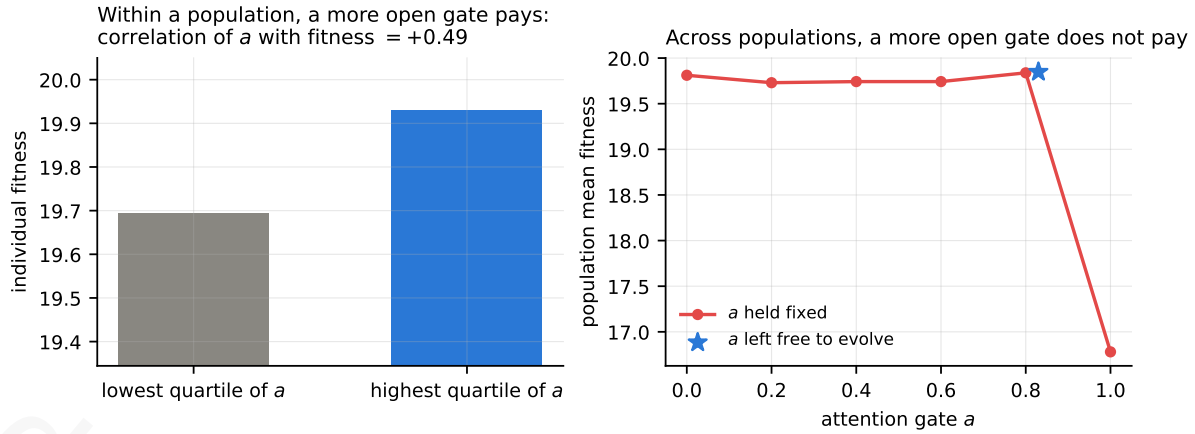


Figure 3: Left: within a population with a evolving, individuals with a more open gate earn more fitness (the vertical axis does not start at zero). Right: across populations with a held fixed, mean fitness is flat until the channel is fully open and collapses at $a = 1$; the star is the population left free to evolve a.

Table 9: Population mean fitness with a held fixed, same conditions as Table 8. Bold marks the collapse at $a = 1$.

a (fixed)	0.0	0.2	0.4	0.6	0.8	1.0
mean fitness	19.811	19.730	19.742	19.742	19.838	**16.781**

is positive and not small: the most socially attentive quarter of the population outperforms the least attentive quarter by about a fifth of a fitness point. Selection accordingly pushes a up to 0.83, short of the collapse at $a = 1$. But holding a fixed across whole populations shows the opposite at the extreme: a population held at $a = 1$ earns three fitness points less than one held at $a = 0$, because a population in which nobody guesses independently has no new information for the social channel to carry. This is Rogers' [29] paradox – a trait that is individually advantageous to adopt can leave the population no better off, or worse off, than not having it – and nothing in equation (2) encodes it; it falls out of letting a evolve on a landscape where information has to be produced by someone before it can be copied by anyone. The population that evolves a freely reaches 19.85, as high as the best fixed setting in Table 9, which is the standard escape from the paradox [49, 32]: such a population stays polymorphic, with low-a individuals continuing to generate the information that high-a individuals free-ride on, rather than collapsing onto the individually-favoured extreme.

5 Discussion

The social channel in this model is permissive, and the analysis says in what sense. The change in expected fitness when one more locus is left undetermined, equation (9), has one term that rewards plasticity, and it is proportional to the rate of environmental change; the channel parameters appear only in the term that charges for it. In a static environment the reward is zero, and no setting of the channel makes selection favour plasticity: the channel weakens the penalty by more than an order of magnitude and at full strength neutralises it, but it never reverses it. When the environment changes, selection on plasticity reverses sign at a threshold that the channel lowers and cannot cross alone, and without the channel the population collapses before the threshold is reached. In the vocabulary of developmental biology, environmental change is the instructive cause of selection for plasticity and social learning is a permissive one: necessary,

dose-dependent, and silent about direction.

This bears directly on the debate with which we began. The disagreement between the extended and the standard synthesis over plasticity-led evolution is in large part a disagreement about causal role, whether plasticity directs adaptive evolution or only permits it [3, 5, 7]. The model does not settle that question for nature. It shows that the question can be settled by measurement in a case where it is posed exactly, and it returns an answer that gives each side its due. The direction of selection on plasticity is set by the environment, as the standard account holds. The channel through which socially acquired information arrives is a condition without which that direction is never expressed, and whose strength sets how much environmental change is needed before it is, which is what the extended account claims for inclusive inheritance. Permissive is not dispensable. That environmental change is what makes plasticity worth paying for is classical quantitative-genetic ground [37, 50, 13]; what the decomposition adds is where a social channel can, and cannot, enter that account.

The methodological consequence is the sharpest. A high terminal level of plasticity is compatible with strong positive selection and with no selection at all, and in this model it is the latter: what an open channel preserves in a static world is drift under a relaxed penalty, with no locus held at an intermediate frequency. Any claim of the form "plasticity was maintained, therefore plasticity was favoured" needs a direct measurement of selection or a convergence test, and both are cheap; the selection gradient and the drift test used here are the instruments that Levis and Pfennig [5] ask for in natural populations. The same caution applies retrospectively to the computational literature, including Belew's [17] reading of a raised plasticity as selection for learning and the conclusions of Le and colleagues [19]. It also dissolves an old disagreement in that literature: whether social learning alone can guide a population to its target [18, 19] turns, in the present model, not on the learning rule but on whether the demonstrator sample carries variance of its own, a single pool-sampling parameter that neither study varied (Appendix A).

Two further readings follow. First, the split of the channel into a peripheral gate and a central transform gives Heyes's account [25, 33] a form a model can test rather than assume, and the test comes out asymmetric in the way her account expects: the peripheral parameter is necessary, since closing it is lethal under environmental change at every level of plasticity we tried, while the central parameter only deepens the discount and rescues a fraction of otherwise doomed populations. Second, what survives of the Baldwin effect here is a claim about persistence, not speed. The canonical reading of H&N is that learning finds the target faster. Under environmental change the asocial population is not slower, it has collapsed; the channel does not accelerate the search so much as keep a population alive long enough for selection on plasticity to become positive at all. That is closer to Baldwin's own pairing of organic selection with social heredity [11] than to the search-efficiency reading: the means of staying in the game are socially supplied, and the reasons to change are not.

What the exercise offers the wider debate is less a verdict than a protocol. Any claim that an inclusive-inheritance channel, social learning, parental effects, epigenetic marks, leads or merely accompanies adaptive evolution can be put in the form used here: write expected fitness as a function of the trait, ask which term of the resulting decomposition the channel enters, and measure the selection gradient under interventions on the channel and on the environment separately. Three extensions seem most informative. First, landscapes with structure, where learning has a gradient of its own, test whether the permissive verdict is a property of social information or of the needle. Second, the predicted scaling of the threshold with L/G, and the effect of a Mayley-type cost of learning, are direct analytic targets. Third, and most pointedly for the extended synthesis: niche construction is the one channel that could enter the benefit term, because what is transmitted then alters the environment itself, the sole source of that term here. Whether a constructed niche can make socially inherited information instructive, in exactly the sense measured in this paper, is the natural next experiment.

5.1 Limitations

The landscape is a needle, chosen because it is the canonical case and because it has no gradient to confound the measurement. Nothing here shows that the result transfers to landscapes with structure, and the permissive reading may well be an artefact of a fitness function whose only signal is exact match. The target moves by single-locus flips at a constant rate; correlated or autocorrelated change is not tested. Fitness is assessed once, at the end of life, which as Best [18] notes gives the learner the full benefit of learning at no cost, so this model has no cost of learning in Mayley's [38] sense and cannot speak to genetic assimilation driven by such a cost. The social signal is derived from successful individuals, which is a strong selectivity assumption; and the thresholds of Table 3 are specific to $L = 20$, $G = 1000$ and one flipped locus per change event. We have not varied L or G. Equation (10) is a one-generation, mean-field criterion evaluated at the population's mean plasticity, which is why it misses the sign at the boundary. The fitness-informed sample we use to stand in for Best's [18] payoff-proportional selection of models – drawing uniformly from the fittest tenth rather than weighting by fitness continuously – is a tractable approximation to his procedure, not a literal replication of it.

Finally, the peripheral and central labels are ours, applied to a and b by argument rather than by measurement. A referee is entitled to hold that a conformity transform is not obviously more central than an attention gate, and we would not resist the point. What the model establishes is that the two parameters are separable and that they do different work; the mapping onto cognitive architecture is an interpretation.

Data accessibility

All simulation code, the archived outputs behind every number reported, and the scripts that generate each figure are publicly available at `https://github.com/drnamlabs-gif/forces-at-hand`; an archived copy with a DOI will be deposited at Zenodo on acceptance. Archived outputs were produced with Python 3.13 and NumPy 2.4.3 under fixed random seeds. Appendix A contains the demonstrator-sampling comparison referred to in the text.

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Ethics

This study is entirely computational and involved no human participants, animal subjects or fieldwork.

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Competing interests

The author declares no competing interests.

A Can the social channel alone sustain adaptation? The demonstrator sample decides

This appendix reports the experiment, referred to in the main text, that resolves the disagreement between Best [18] and Le [19] over whether social learning alone can guide a population to its target. It uses the second implementation of the model (Section 4): learning budget $G = 300$, k demonstrators sampled per learner, fidelity $f = 1$, $b = 0$, initial $p(?) = 0.5$, $N = 1000$, $L = 20$, 200 generations, 12 independent runs per condition, NumPy 2.4.3 with fixed random seeds.

Best's report [18] that social learning alone finds the target, and Le's report [19] that it does not, were never run against each other on the same code, with everything but the channel held fixed. We do that here, inside equation (2), by fixing $a = 1$, $b = 0$ – learners guess purely from the social signal, with no individual contribution at all – and varying only the sampling rule for c_j.

Table A1: Seven regimes on the static landscape, $L = 20$, $G = 300$, $N = 1000$, 200 generations, 12 runs. SWITCH is the original mechanism of Le [19], discontinuous mechanism (oracle-gated copying), re-implemented as a separate baseline rather than a point in equation (2).

configuration	mean F	final $p(?)$	final $p(1)$	solved / 12	first solved (gen.)
evolution alone (no plasticity)	4.16	0.000	0.572	2	144
individual learning alone, $a = 0$	19.69	0.094	0.906	12	0
social learning alone, k=1, parent	**1.00**	0.515	0.237	**0**	–
social learning alone, k=5, fittest tenth	**16.78**	0.482	0.470	**10**	8
blend, $a = 0.8$, $k = 5$, fittest tenth	18.26	0.455	0.519	12	8
blend, a evolves, $k = 5$, fittest tenth	19.83	0.414	0.586	12	2
SWITCH (original mechanism of [19])	19.33	0.530	0.470	12	0

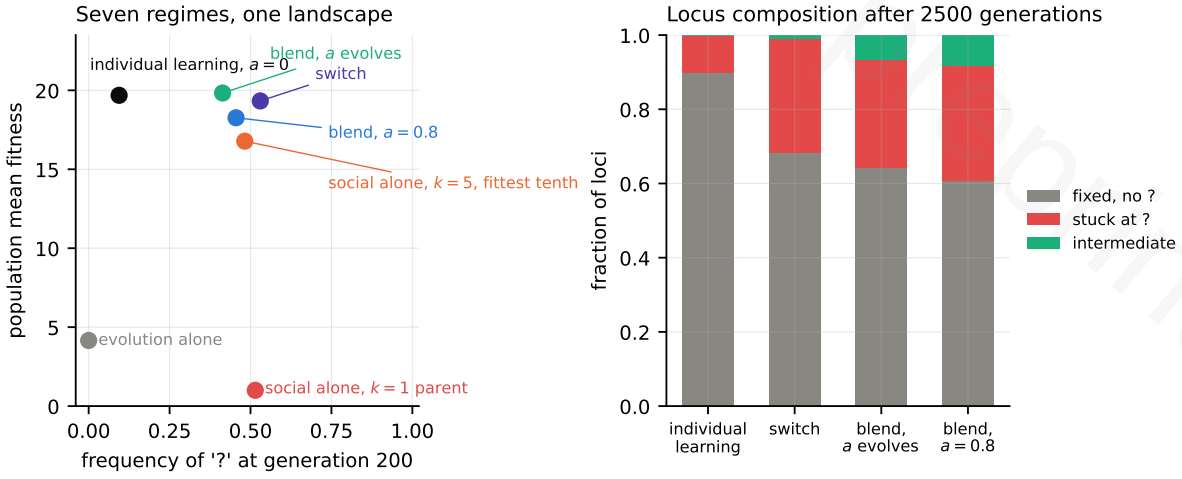


Figure A1: The seven regimes of Table A1 on the (plasticity, fitness) plane, and the locus composition each reaches after 2500 generations (Table A2).

The middle two rows differ only in k and the sampling rule, with $a = 1$, $b = 0$, fidelity, cost and everything else held identical. $k = 1$ under PARENT reproduces Le's result: not one of twelve runs finds the target. $k = 5$ drawn from the fittest tenth reproduces Best's: ten of

twelve runs find it, with a median first solution at generation eight. The two reports that stood as a contradiction for twenty years are two points on one axis, **whether the demonstrator sample carries variance**. A single inherited demonstrator is a conduit: it transmits a value but manufactures none, so a population without individual guessing has no source of new information and cannot discover anything it has not already discovered. A sample of five drawn from a fitness-selected pool is a search process in its own right, with variance across generations in who belongs to the pool; that variance supplies exactly the exploration that a single demonstrator cannot, which is sufficient, on this landscape, to find the target with no individual guessing at all. The statement "social learning cannot learn anything that has not already been learned" is therefore true of the $k = 1$ row and false of the $k = 5$ row, and the field's disagreement was over which row its intuitions were about.

Table A2 extends all four plastic regimes to 2500 generations.

Table A2: After 2500 generations. "Stuck" is the fraction of loci still undetermined; "free" is the fraction fixed to a determined value; "mid" is the fraction at an intermediate frequency across the population.

configuration	population $p(?)$	free	**stuck**	intermediate
individual learning alone	0.100	0.900	**0.100**	0.000
SWITCH [19]	0.310	0.683	**0.308**	0.008
blend, a evolves	0.327	0.642	**0.292**	0.067
blend, $a = 0.8$	0.354	0.608	**0.308**	0.083

Every regime with a social channel converges to roughly three times the asocial stuck-locus fraction, whichever mechanism supplies the channel, and the population-level frequency of "?" tracks the stuck fraction almost exactly in every row. Section 4.3 argues this equality is itself diagnostic.

This has a direct consequence for how earlier results, including Le's [19], should be read. A run that ends with half its loci undetermined, having been initialised with half its loci undetermined, is not evidence that genes have stopped assimilating because plasticity became valuable. It is evidence that selection stopped discriminating. The distinction is invisible to the frequency diagnostic and immediate under β_q.

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